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Implementing Scrum Using Team Foundation Server 2010

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Good afternoon

Introductions are in order



Richard Hundhausen

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Accentient

- ▶ A leader in Visual Studio ALM knowledge
- ▶ Has helped hundreds of clients understand, implement and leverage Team Foundation Server
- ▶ Has a close working relationship with Microsoft
- ▶ Has a cross-functional team consisting of Microsoft MVPs, Microsoft RDs, Certified Professional Scrum Developers, Certified Scrum Professionals, and Microsoft Certified Trainers

www.accentient.com

The Professional Scrum Developer Program



Finally – training, assessment, and certification for the *team*

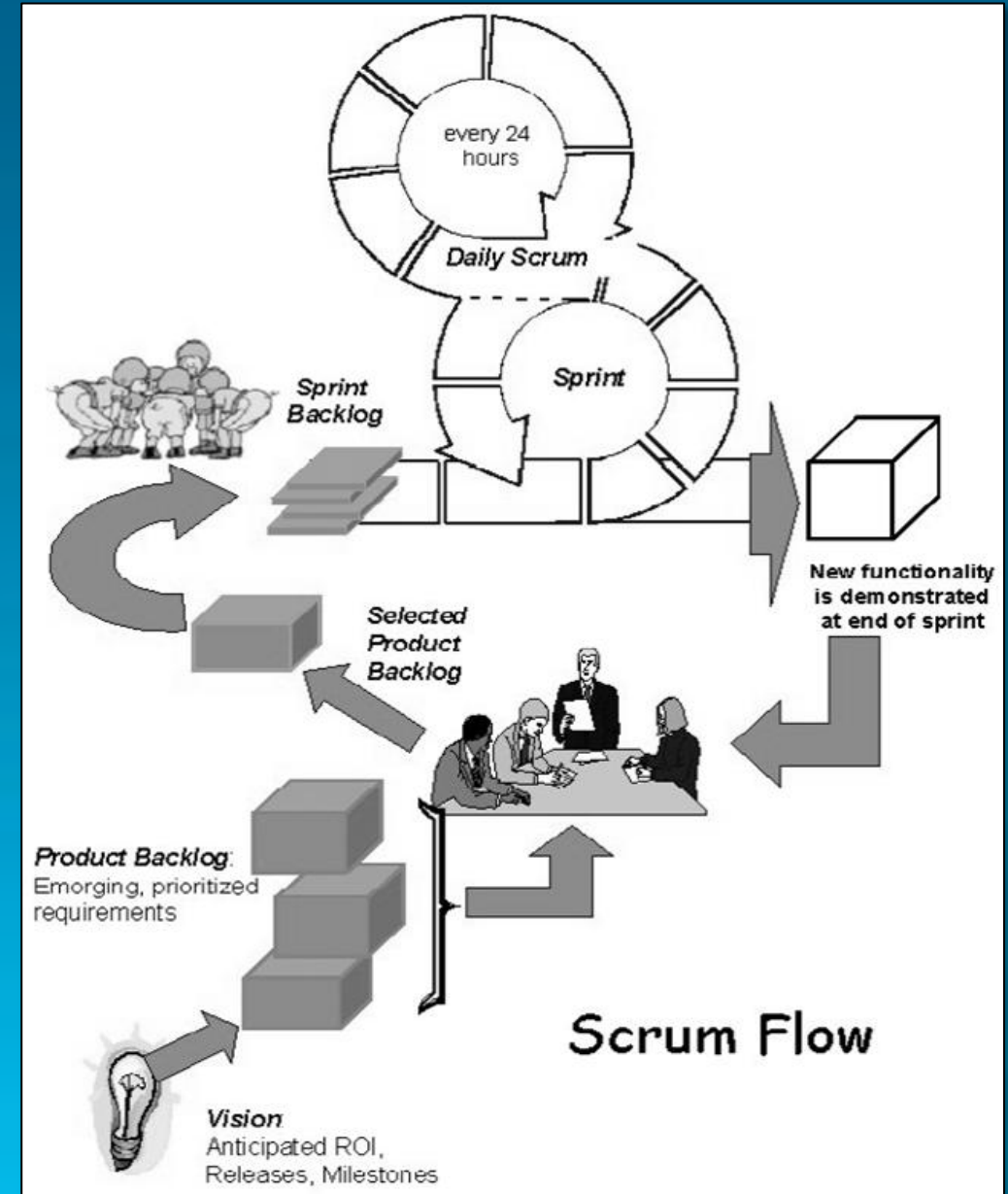
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Visual Studio Scrum



Review: Scrum on a Slide

- ▶ Roles
 - ▶ Product Owner, ScrumMaster, Team
- ▶ Timeboxes (a.k.a. ceremonies)
 - ▶ Release Planning meeting
 - ▶ Sprint Planning meeting
 - ▶ The Sprint
 - ▶ Daily Scrum
 - ▶ Sprint Review meeting
 - ▶ Sprint Retrospective meeting
- ▶ Artifacts
 - ▶ Product Backlog
 - ▶ Sprint Backlog
 - ▶ Burndown charts
 - ▶ **Working Software!**



Microsoft Visual Studio Scrum 1.0

- ▶ A process template introduced shortly after Visual Studio 2010 launched
 - ▶ This demonstrates how popular Scrum is and also Microsoft's commitment to supporting Scrum
- ▶ Maps directly to the Scrum concepts
 - ▶ Sprint vs. Iteration, PBI vs. User Story, Impediment vs. Issue
- ▶ Available as free download from Microsoft:
 - ▶ <http://bit.ly/chyBhE>

Visual Studio Scrum 1.0 Artifacts

Work Item Types	Team Queries	Reports
Sprint	All Sprints	Release Burndown
Product Backlog Item	Product Backlog	Sprint Burndown
Task	Blocked Tasks	Velocity
Impediment	Open Impediments	Build Success Over Time *
Bug	Sprint Backlog	Build Summary *
Test Case *	Test Cases	Test Case Readiness *
Shared Steps *	Unfinished Work	Test Plan Progress *
	Work in Progress	

* Nearly identical to that found in the MSF for Agile Software Development 5.0 template

Mapping Scrum to Visual Studio Scrum 1.0 *

Scrum Concept	Visual Studio Scrum 1.0
Team	Team Foundation Server security groups
Definition of Done	Wiki entry or document on the portal
Product	Team Project
Product Backlog	Product Backlog team query
Product Backlog Item	Product Backlog Item work item
Acceptance Criteria	Acceptance Criteria field and (optionally) associated Test Case work items
Release	Iteration Path (high level)
Release Goal	Wiki entry or document on the portal
Release Capacity	Wiki entry or document on the portal
Sprint	Iteration Path (low level) and associated Sprint work item
Sprint Dates	Start Date and Finish Date fields
Sprint Goal	Sprint Goal field
Sprint Capacity	Wiki entry or document on the portal
Sprint Backlog	Sprint Backlog team query
Task	Task work item
Burndown and Velocity	Release Burndown, Sprint Burndown, and Velocity reports
Impediment	Impediment work item
Bug	Bug work item
Retrospective	Retrospective field in the Sprint work item

DEMO

The Product Backlog



Product Backlog Item Work Item

The screenshot shows a 'New Product Backlog Item' form. Annotations with leader lines point to various fields:

- Specify title:** Points to the 'Title' field, which currently contains '<Required>'.
- Specify iteration path:** Points to the 'Iteration' dropdown menu, which is set to 'Team Project'.
- Specify relative priority, effort, and business value:** Points to the 'Backlog Priority' (1000), 'Effort', and 'Business Value' fields.
- Specify owner:** Points to the 'Assigned To' dropdown menu.
- Link to tasks:** Points to the 'Tasks' tab in the bottom section.
- Link to test cases that test the backlog item:** Points to the 'Test Cases' tab in the bottom section.
- Attach files:** Points to the 'Attachments' tab in the bottom right section.
- Specify the criteria of completing the backlog item:** Points to the 'Acceptance Criteria' tab in the bottom right section.

The form includes a 'Save Work Item' button and a warning message: 'New Product Backlog Item 1 : TF20012: Field 'Title' cannot be empty.' The bottom section contains tabs for 'Description', 'Test Cases', 'Tasks', 'Acceptance Criteria', 'History', 'Links', and 'Attachments'. The 'Description' tab is active, showing a template: 'As a <type of user> I want <some goal> so that <some reason>'.

Bug Work Item Type

The screenshot shows a 'New Bug' form with the following fields and annotations:

- Title:** An empty text field. Annotation: "Specify title".
- Iteration:** A dropdown menu with "Team Project" selected. Annotation: "Specify owner".
- Status:** A dropdown menu with "New" selected.
- Assigned To:** A dropdown menu.
- State:** A dropdown menu with "New" selected.
- Reason:** A dropdown menu with "New defect reported" selected.
- Backlog Priority:** A text field with "1000" entered. Annotation: "Specify relative backlog priority".
- Effort:** A text field.
- Severity:** A dropdown menu with "3 - Medium" selected. Annotation: "Specify relative effort".
- Area:** A dropdown menu with "Team Project" selected. Annotation: "Select the level of severity".
- Link to test cases and tasks:** A button above the "Steps to Reproduce" and "Acceptance Criteria" sections.
- Steps to Reproduce:** A text area with a rich text editor toolbar. Annotation: "Specify build number and system information".
- Acceptance Criteria:** A text area with a rich text editor toolbar. Annotation: "Specify the expected behavior".
- Attachments:** A section for attaching files. Annotation: "Attach files to support bug resolution".

At the bottom left, a general instruction states: "Add information that is required to reproduce the bug".

Specifying Acceptance Criteria

- ▶ This can be done incrementally

Disambiguity

1. Initially, a PBI can just have a title
2. Later, the PBI can be updated to include a more detailed description
3. Later, the PBI can be updated to include detailed acceptance criteria
4. Later, the PBI can be linked to a Test Case work item with more detailed criteria in its description field
5. Later, the Test Case can have detailed steps specified (MTM only)
6. Manual tests can be recorded and played-back as automated tests (MTM only)

The Sprint



Activities: Setting up a New Sprint

- ▶ Ensure that the release and Sprint nodes exist
- ▶ Ensure there is a corresponding Sprint work item
- ▶ Modify the Current Sprint queries to reference the current iteration (Sprint)
- ▶ Run the Product Backlog query for accuracy
- ▶ Review the Release Burndown and Velocity reports
- ▶ Edit the PBI work items you are committing to deliver
- ▶ Create and link initial task work items
- ▶ Review the Sprint Backlog query for accuracy

Sprint Work Item Type

Specify the iteration path of the sprint

Review the success and failure of the sprint

New Sprint 21* X

Save Work Item

New Sprint 21

Iteration: Team Project

Details Retrospective

Start Date: ☐ ▼

Finish Date: ☐ ▼

Sprint Goal

Specify goals for the sprint

History Attachments

B I U A

Sprint Backlog

- ▶ The Sprint Backlog is a list of tasks that defines a team's work for a Sprint
 - ▶ Tasks emerge during Sprint planning
- ▶ The Sprint Backlog is what the Team must complete to meet the Sprint Goal
- ▶ The Sprint Burndown chart is based on the Sprint Backlog being accurate and up-to-date

Activities: Daily Scrum Tasks

- ▶ Create and link task work items as needed
- ▶ Team takes on (signs up) for tasks from the Sprint Backlog
- ▶ Check-in any code changes to Team Foundation Server associating with a single task only
- ▶ Update remaining work estimates on your tasks
- ▶ Complete your work according to the acceptance criteria and your team's Definition of Done
- ▶ As you complete a task, set the state and remaining work
- ▶ Update the Product Backlog by creating and editing PBI and Bug work items as needed (grooming)
- ▶ Create and manage impediment work items as needed
- ▶ Review the Sprint Burndown report to monitor progress

Task Work Item Type

The screenshot shows a 'New Task' form with various fields and sections. Annotations with leader lines point to specific elements:

- Specify title**: Points to the 'Title' text input field.
- Specify iteration path**: Points to the 'Iteration' dropdown menu, which currently shows 'Team Project'.
- Specify owner**: Points to the 'Assigned To' dropdown menu.
- Estimate remaining work**: Points to the 'Remaining Work' text input field.
- Specify relative backlog priority**: Points to the 'Backlog Priority' text input field.
- Specify activity type**: Points to the 'Activity' dropdown menu.
- Provide additional information**: Points to the 'Description' text area.
- Indicate whether the task is currently blocked**: Points to the 'Blocked' dropdown menu.
- Attach files**: Points to the 'Attachments' tab and its associated text area.
- Link to other work items**: Points to the 'Links' tab and its associated text area.

The form includes a 'Save Work Item' button and a warning message: 'New Task 22 : TF20012: Field 'Title' cannot be empty.' The 'State' is set to 'To Do' and the 'Reason' is 'New task'. The 'Area' is also set to 'Team Project'.

Test Case Work Item Type

The screenshot displays the 'New Task Case 1' form. At the top, a yellow banner contains the text 'New Test Case 1 : TF20012: Field 'Title' cannot be empty.' Below this, the form is divided into several sections. The 'Title' field is labeled '<Required>'. The 'Status' section includes 'Assigned To: Andrew Hill', 'State: Design', 'Priority: 2', and 'Automation Status:'. The 'Classification' section has 'Area: Dev10Demo' and 'Iteration: Dev10Demo'. A button labeled 'Click to modify steps in Test Manager' is visible. Below the form, there are tabs: 'Steps', 'Summary', 'Tested User Stories', 'All Links', 'Attachments', and 'Associated Automation'. The 'Steps' tab is active, showing a table with columns 'Action' and 'Expected Result'. The table contains two rows: '1. Login to the application' and '2. Click the File menu.'. Annotations with arrows point to various parts of the form: 'Add title' points to the title field; 'Select owner of test case' points to the 'Assigned To' field; 'Select Planned if test is automated' points to the 'Automation Status' field; 'Select the area and iteration paths' points to the 'Area' and 'Iteration' dropdowns; 'Click to modify steps in Test Manager' points to the corresponding button; 'Attach files' points to the 'Attachments' tab; 'Add test method to automate the test case' points to the 'Expected Result' column; 'Link to other work items, changesets, versioned items, or URLs' points to the 'All Links' tab; 'Link to backlog items and bugs' points to the 'Tested User Stories' tab; and 'Add more details and view history' points to the 'Summary' tab.

Annotations:

- Add title
- Select owner of test case
- Select Planned if test is automated
- Select the area and iteration paths
- Click to modify steps in Test Manager
- Attach files
- Add test method to automate the test case
- Link to other work items, changesets, versioned items, or URLs
- Link to backlog items and bugs
- Add more details and view history

Impediment Work Item Type

The screenshot shows the 'New Impediment' form in Jira. The form is titled 'New Impediment 19*' and has a 'Save Work Item' button. A yellow error message states: 'New Impediment 19 : TF20012: Field 'Title' cannot be empty.' The form fields are as follows:

- Title:** A text field that is currently empty. An annotation 'Add title' points to this field.
- Iteration:** A dropdown menu with 'Team Project' selected. An annotation 'Specify iteration path' points to this dropdown.
- Status:** A dropdown menu with 'New' selected.
- Assigned To:** A dropdown menu.
- Priority:** A dropdown menu with '2' selected.
- Area:** A dropdown menu with 'Team Project' selected.
- Reason:** A dropdown menu with 'New impediment' selected.

Below the form fields are three tabs: 'Description', 'Resolution', and 'History'. The 'Description' tab is active, showing a text area with a rich text editor toolbar. An annotation 'Provides additional information' points to the text area. The 'Resolution' tab is also visible, showing a text area with a rich text editor toolbar. An annotation 'Describe resolution' points to the text area. The 'History' tab is also visible, showing a text area with a rich text editor toolbar. An annotation 'Link to other work items' points to the text area. The 'Links' and 'Attachments' tabs are also visible, showing text areas with rich text editor toolbars. An annotation 'Attach files' points to the text area.

Fix Bugs, Don't Manage Them

- ▶ For PBIs you are currently working on (“in Sprint bugs”) ...
 - ▶ If it's a small bug ($< n$ hours to fix), then just fix it
 - ▶ If it's a larger bug ($> n$ hours to fix) and won't impact the ability to achieve the Sprint goal, then create a Bug work item, associate a Task and have a team member perform the fix during the Sprint
 - ▶ If it's a larger bug ($> n$ hours to fix) and will impact the ability to achieve the Sprint goal, then create a Bug work item to be prioritized (by the Product Owner) and to be fixed in a later Sprint; you may not be able to achieve your Sprint goal
 - ▶ If another team member finds the bug, then they can create a Bug work item and then the team decides if it can be fixed in the current Sprint or needs to be prioritized (by the Product Owner) to be fixed in a later Sprint, in which case you may not be able to achieve your Sprint goal

Fix Bugs, Don't Manage Them

- ▶ For PBIs you thought were *done* (“out of Sprint bugs”) ...
 - ▶ If the bug is not critical (i.e. a hotfix), then create a Bug work item to be prioritized and fixed in a later Sprint
 - ▶ For a critical hotfix (Product Owner decides), do whatever needs to be done to get the fix into production, knowing that your current Sprint commitments may be impacted
 - ▶ Adjust the team's capacity accordingly, especially if lots of hotfixes start occurring
- ▶ General tip: don't create an associated task to fix a bug until the Sprint in which the team commits to fix it

Activities: Completing a Sprint

- ▶ Label code and other artifacts in version control
- ▶ Set the State of the PBI and/or Bug work items to Done when the Product Owner accepts it
- ▶ Set the State of the PBI/Bug and linked work items appropriately for unfinished/rejected work
- ▶ Update the Product Backlog by creating and editing PBI and Bug work items as needed
- ▶ Capture the Sprint Retrospective feedback on the Sprint work item
- ▶ Branch as needed (i.e. to support hotfixing)

Getting to Done

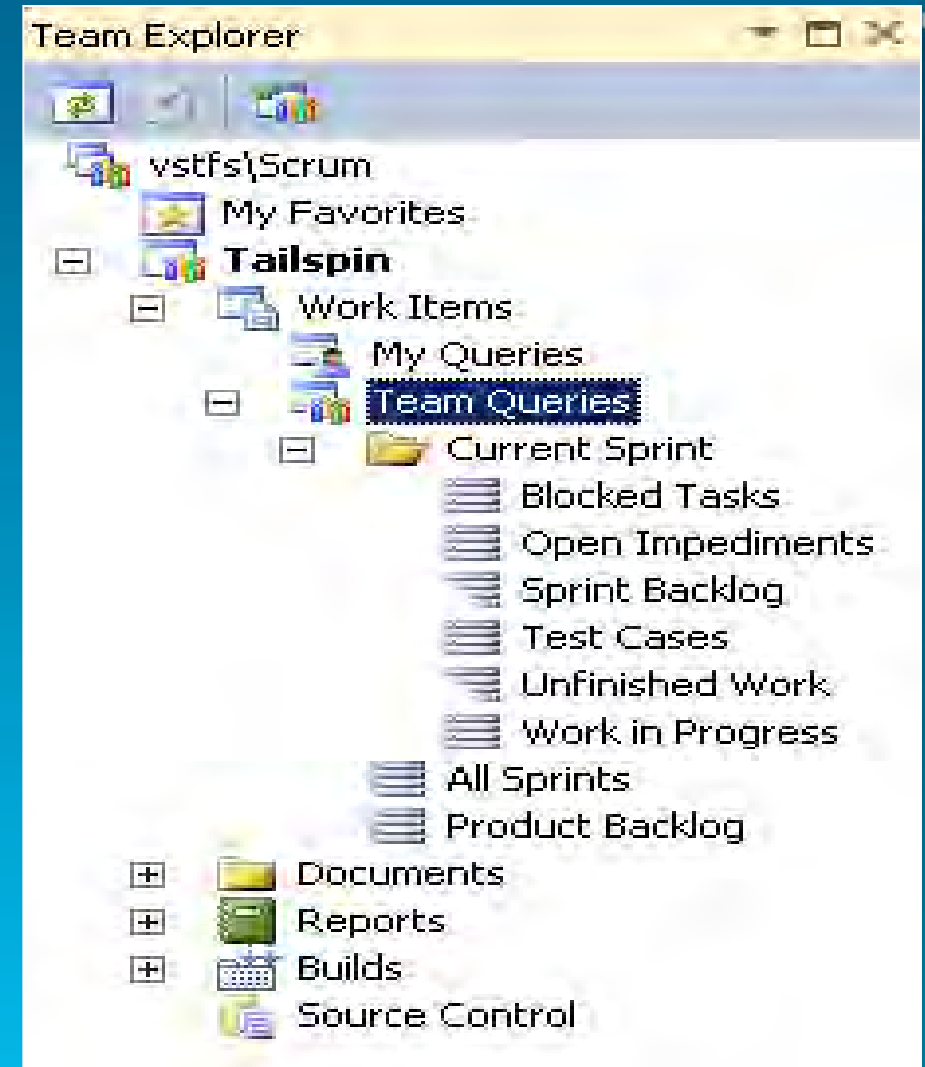


"Done"

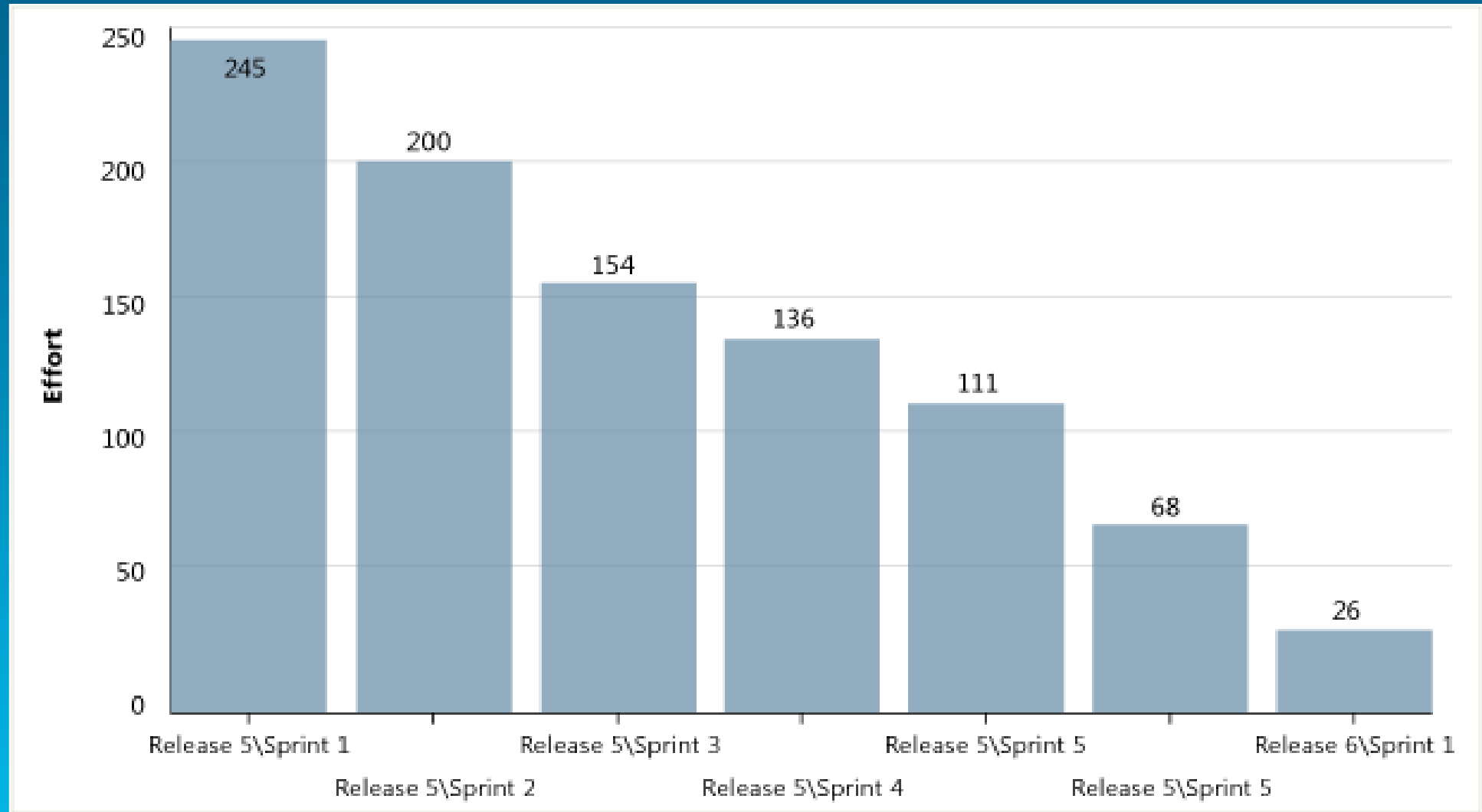
- ▶ Done defines when an increment of product functionality is potentially shippable
- ▶ Definition of Done (DoD)
 - ▶ A simple, auditable checklist owned by the team
 - ▶ It can be influenced by organizational standards and specific requirements of the product or release
- ▶ Here is a simple Definition of Done
 1. Code complete
 2. Tests pass
 3. Everything checked-in
 4. Acceptance criteria from Product Owner met

Queries

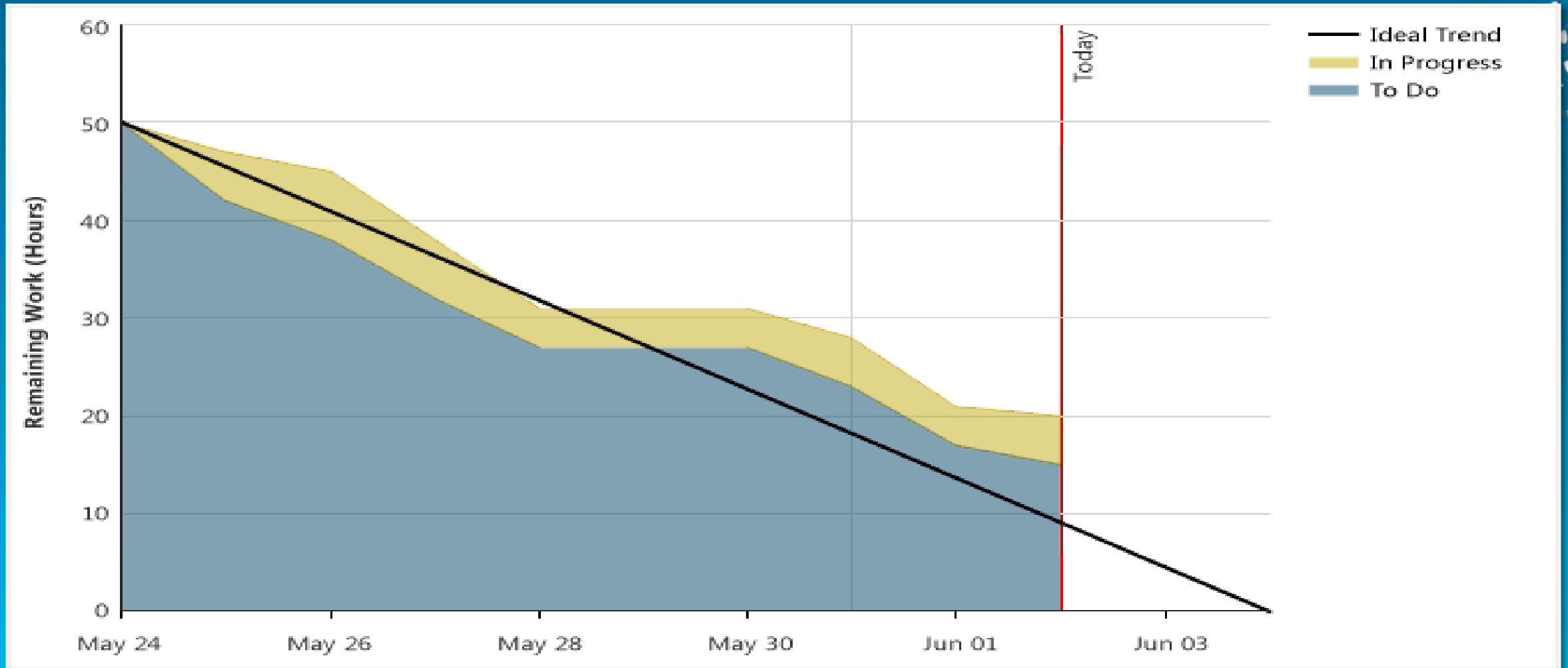
- ▶ Product queries
 - ▶ All Sprints
 - ▶ Product Backlog
- ▶ Current Sprint queries
 - ▶ Blocked Tasks
 - ▶ Open Impediments
 - ▶ Sprint Backlog
 - ▶ Test Cases
 - ▶ Unfinished Work
 - ▶ Work in Progress
- ▶ Others ...



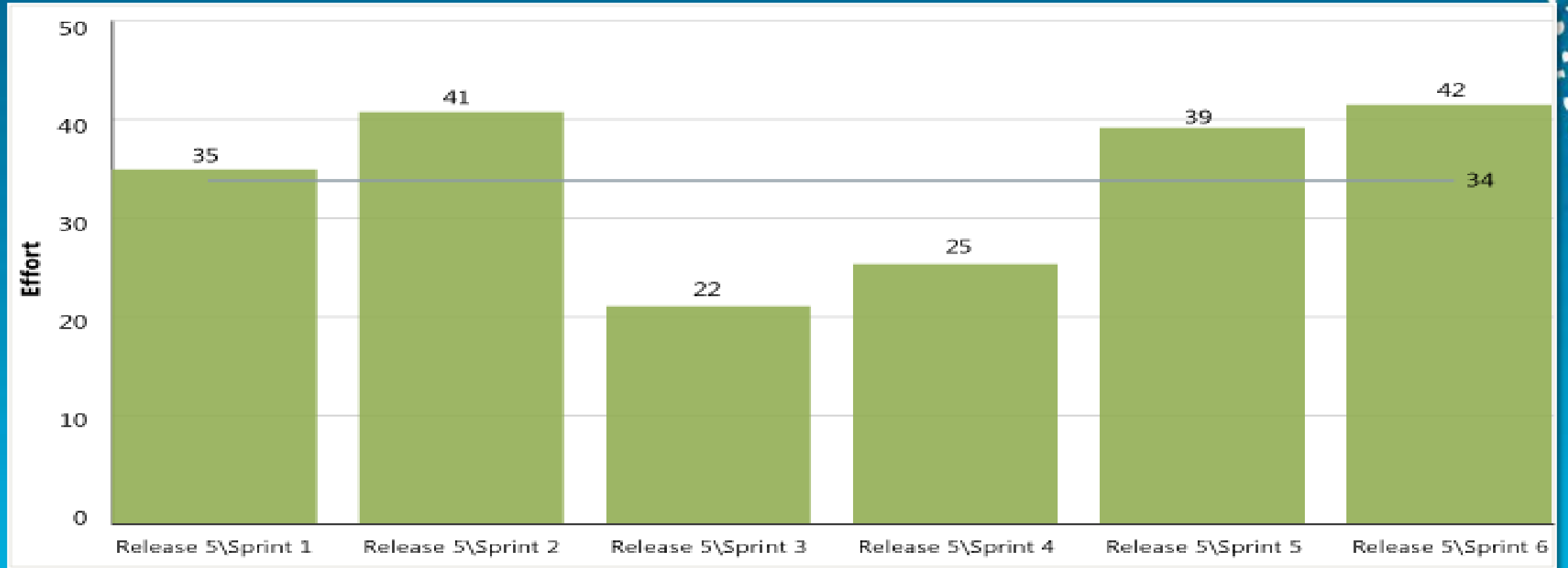
Release Burndown Report



Sprint Burndown Report



Velocity Report



Overcoming Dysfunction



The Team

- ▶ Responsible for *committing* to work
- ▶ Cross-functional
- ▶ Self-organizing
- ▶ Ability to do whatever is needed to meet commitment
- ▶ Adheres to professional ethics
- ▶ Demonstrates Sprint output

Missing a Key Team Member

As a team, what do you do when Simon goes missing?

Simon is the application architect and the only one with the vision of what the software should do. Something has happened in his personal life and he's left for Australia and gone walkabout. Nobody knows how to reach him or if/when he'll return.

Four Developers = $\frac{1}{4}$ work each?

As a team, how would you handle this?

The team consists of four senior developers. They want to take $\frac{1}{4}$ of the selected product backlog each. They say they will pull it together in the last several days of the Sprint. Because of this they say that daily Scrums aren't needed.

What about Laura?

As a team, how would you handle Laura?

Laura is an excellent C# programmer. She says that it's the job of QA to test, so she is not going to write unit tests.

What about Dieter?

As a team, how would you handle Dieter?

Dieter is your organization's only SQL Server administrator. He holds all the passwords and all database changes must go through him. He's happy to help your team with its database tasks, but only for a few days. He's very busy, and important. He makes it clear that he doesn't work for you or your boss and his help is just a courtesy.

Final Exam: What if you are not really done?

As a team, how would you handle this situation?

Your team believes that they have met their Sprint Goal and the Product Owner is quite happy with the increment's functionality. One of the PBIs wasn't completed according to your definition of done. It seems that a Code Analysis warning remains:

Error List			
 0 Errors  1 Warning  0 Messages			
		Description	Project
	1	CA2210 : Microsoft.Design : Sign 'Tailspin.Admin.App.exe' with a strong name key.	Tailspin.Admin.App

Done



Learn more here: www.scrum.org

Check out our training services: www.accentient.com

Contact me: richard@accentient.com